

Its downsides include non-continuous sampling over large intervals, and the large potential for human error. Additionally, volatile substances may be initially captured as fine particulates, but they have the ability to sublime off of the filters, and are thus only intermittently recorded.

2.2.2 GRIMM Laser Aerosol Spectrometer (LAS)

The GRIMM EDM 180 is a federal equivalence monitor that calculates mass at multiple size cutoffs by counting and sizing particles optically [33]. The GRIMM is capable of discriminating between 32 distinct size ranges of particles. Because the measured particles are categorized by size, no cyclones or impactors are required; only very coarse filtering to prevent bugs and singly-visible particles from entering. Because the instrument is based on optical measurement, however, the device must account for or reduce hygroscopic growth due to humidity. The GRIMM accomplishes this by transporting newly sampled air down through a long nafion tube. Dry, non-sample air flows upward in a concentric pipe around the nafion tube, extracting moisture from the intake air in the process. An additional concentric pipe recirculates the air back into the base unit for expulsion. The pressure needed to circulate this air necessitates a large and relatively noisy pump. The pump only receives power at an observed humidity of around 70%.



Figure 2.1: Rack-mount GRIMM EDM 180 and sampling tube [3]

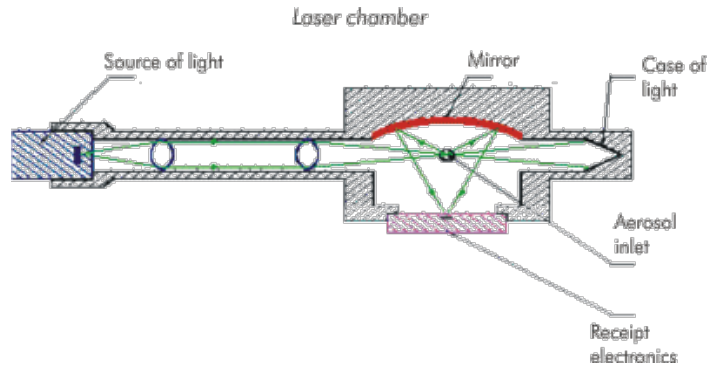


Figure 2.2: Schematic of the laser chamber in the GRIMM Model 1.109 LAS [3]

2.2.3 Beta Attenuation Monitor (BAM)

Beta attenuation monitors (BAMs) operate on the principle that beta radiation from any particular source is attenuated by obstructing solid matter. Typically, a continuous ribbon cycles past a window of controlled airflow. The ribbon is made of fine filter paper capable of capturing $PM_{2.5}$ matter. Two detectors on either side of the window measure the beta attenuation level before and after that section of ribbon is exposed to the airflow, and the difference in attenuation can be used to calculate the increase in mass of the ribbon [19]. Given a known flow rate and a known exposure time, airborne particle concentration is easily determined.

Some shortcomings of this method include the necessity to pre-filter the air to the appropriate size particle distribution (usually $PM_{2.5}$ or PM_{10}) using a cyclone assembly. The minimum sampling rate is typically once per hour, whereas optical or resonant methods can report samples every minute (though shorter sampling times lead to greater error, especially at low concentrations). Additionally, while the GRIMM is capable of delivering estimates for $PM_{2.5}$ and PM_{10} simultaneously, this is not possible with the BAM. The BAM also suffers from the volatile sublimation issue present in the FRM.

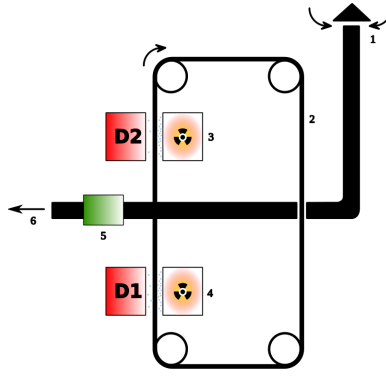


Figure 2.3: Diagram of BAM operation [41]

2.2.4 Tapered-Element Oscillating Microbalance (TEOM)

TEOMs operate by directing particles from an airstream onto or through a small filter or plate attached to a thin oscillating beam. The total mass on the plate can be determined by calculating the resonant frequency of the beam and plate [8]. Like the BAM, particle concentration is calculated using the difference in accumulated mass across the sample interval. These intervals can be short (e.g. one minute) but shorter intervals can sometimes result in imprecise readings. This is especially true if a large number of the particles are volatile and evaporate from the plate or filter, as this can cause a net loss of mass across a small time interval, which is then used to calculate an impossible negative particle concentration. Also, similar to the BAM, cyclones and impactors must be used to separate out particles that are larger than the cutoff size of interest.

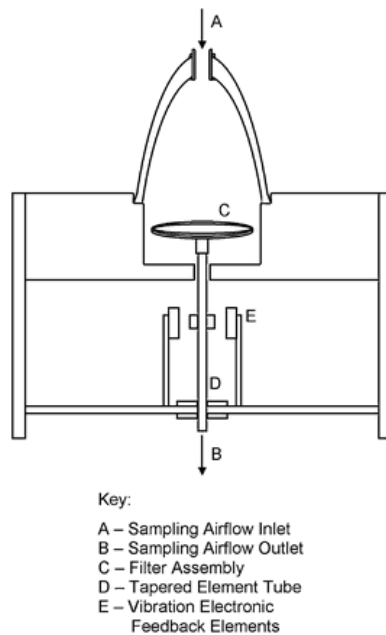


Figure 2.4: Diagram of TEOM operation [38]

2.2.5 Handheld Optical Particle Counter (OPC)

Handheld optical particle counters such as the Hach Met One HHPC-6+ are significantly less expensive and more portable than the previous devices, but are not considered federal reference methods. The principle of operation behind these devices is similar to that of the GRIMM, where a laser is used to count and size particles drawn through the sensing chamber using a small pump. Typically, the sample volume is smaller and little or no preconditioning is performed on the sample air. Temperature and relative humidity are frequently measured in these units at the point of particle detection. Devices like the HHPC-6+ frequently incorporate a basic user interface, on-board storage of varying capacities, and a rechargeable battery. These devices can be very useful for point sampling, but are not designed for continuous operation. Additionally, while some devices like the TSI DustTrak will estimate mass, the HHPC-6+ delivers only particle counts across different size bins, making comparison with mass-only methods like the BAM difficult. If the particle source is well characterized, it is possible to estimate mass with an OPC that possesses at least micron-level size resolution [9].

2.3 Existing Low Cost Air Quality Monitors

Related works [21] have established that inexpensive sensors designed to detect larger dust particles can also be used to detect particles in the $2\mu m$ range. Typically, these experiments use a moving average of raw sensor readings, with a linear scale factor used to match these